🦜 Parrot Assistant v1 — Assembly Card

Tomasz Kolinko · Apr 2026 · full guide at `kolinko.eu/parrot`

① Unboxing checklist

Kamami (order #337481, ~292 PLN)

- ☐ XIAOML Kit · `1202930` · 1×
- ☐ MAX98357A · `1182935` · 1× — verify 6-pin header is pre-soldered by Kamami!
- ☐ Speaker 28 mm · `234920` · 1×
- ☐ JST-PH 2.0 pigtail · `1181279` · 3×
- ☐ Magnets 10×1.5 mm · `1186978` · pack of 10

Botland (order #1618961, ~43 PLN)

- ☐ F-F jumper ribbon · `19620` · 1× (use 6)
- ☐ USB-A→USB-C cable · `14829` · 2×

Printstein (~65 PLN)

- ☐ body_top.stl (matte light grey PLA)
- ☐ belly_plate.stl
- ☐ feet.stl (×2)

User-sourced

- ☐ USB power bank (Wozinsky WPower 10000 or existing)
- ☐ Brooch bar pin ~38 mm (pasmanteria)

② Tools

Required: CA (super) glue · scissors or wire strippers · PVC insulation tape.
Optional: multimeter · Posca paint markers (white/black/red).
Not needed: soldering iron · 3D printer · crimper.

③ Assembly steps

1. **Flash XIAO first** (before gluing) — see back panel.
2. **Glue magnets:** 2 in belly plate (N up/out), 2 in body_top floor (N down/in). CA glue, hold 20 s. *Dry-fit: belly plate should **snap** shut.*
3. **Glue brooch bar pin** into rear recess on belly plate (external). Hold 30 s, cure 10 min.
4. **Friction-fit feet** into belly plate holes (dab CA glue if loose).
5. **Speaker→JST pigtail:** strip 5 mm, twist pairs, wrap with tape. Tug-test.
6. **Wire XIAO ↔ MAX98357A** (6 F-F jumpers, see wiring table).
7. **Plug JST plug** into MAX98357A PH2.0 socket.
8. **Mount boards** in body_top cavity with 2 cm² foam tape (camera bore at front, USB-C at tail).
9. **Drop speaker** into hex grille pocket (belly-facing).
10. **Close belly plate** — magnets snap shut.
11. **Thread USB-C cable** through tail slot into XIAO port.
12. **Plug into power bank** — double-press flashlight button to lock always-on.
13. **Pin** through shirt shoulder seam via brooch bar.

④ Wiring: XIAO ↔ MAX98357A

XIAO	GPIO	MAX98357A	What
3V3	—	VIN	+3.3 V supply
GND	—	GND	ground
GND	—	**SD**	**tie LOW** (enables amp)
D1	2	BCLK	I²S bit clock
D3	4	LRC	I²S word-select
D4	5	DIN	I²S data in

Gotcha: leaving SD floating silently mutes the amp — the #1 "no audio" cause. GAIN pin stays unconnected (default 9 dB is fine).

Part 2 — Flash · Boot · First Chat

⑤ Flashing the XIAO

Do this BEFORE gluing. Confirm the board is alive first.

1. Install PlatformIO: `brew install platformio`
2. Find Mac IP: `ipconfig getifaddr en0` (wired) or `en1` (WiFi)
3. Copy template + edit:

```
cd /Users/kolinko/claude/varia/parrot/firmware
cp src/secrets.h.template src/secrets.h
# edit src/secrets.h:
#   WIFI_SSID, WIFI_PASS
#   MAC_BRAIN_HOST = (IP from step 2)
#   MAC_BRAIN_PORT = 8766
```

4. Plug XIAO via USB-C into Mac. First flash: hold BOOT, tap RESET, release BOOT.
5. Compile + upload:

```
pio run
pio run -t upload
pio device monitor -b 115200
```

6. Look for: `ws: connected` — that's the green light.

⑥ Start the Mac brain

Run preflight first — catches all Mac-side problems before you flash:

```
python3 parrot/brain/preflight.py
```

Want all PASS. Then:

```
cd /Users/kolinko/claude/varia/parrot/brain
./run.sh            # or: ./run.sh dry (no TTS)
```

⑦ First chat

1. Pin parrot to shoulder · plug power bank · lock always-on.
2. Say: "*Hey parrot, can you see me?*"
3. Within ~6–8 s, expect a reply through the speaker.
4. Watch admin panel: `kex7m.effort.ws/parrot`

Expected behaviour:
· **Reactive** = replies when you speak
· **Proactive** = checks every 5 min, talks ≤ once per 20 min
· May say *nothing* intentionally — it's a companion, not a chatbot

⑧ Troubleshooting cheatsheet

Symptom	First thing to try
Power bank auto-shuts	Double-press flashlight button or swap for Voltaic/Green Cell/Anker
WS won't connect	Check MAC_BRAIN_HOST IP `nc -vz MAC 8766` · 2.4 GHz only
No audio out	**SD pin → GND?** (most common)
No audio in	Sense board fully seated? speak louder
Speaker crackles	Seat deeper + foam gasket
Black JPEGs	Peel OV2640 lens film

⑨ Safety

- Indoor only — not waterproof.
- Magnets are strong — away from cards / pacemakers.
- Camera privacy: unplug in confidential rooms (no mic mute in v1).
- LAN-only ws:// — don't take on hotel WiFi.